



JURONG JUNIOR COLLEGE

JC2 Preliminary Examination 2018

CANDIDATE
NAME

CLASS

BIOLOGY

9744/01

Paper 1 Multiple Choice

14 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, class and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

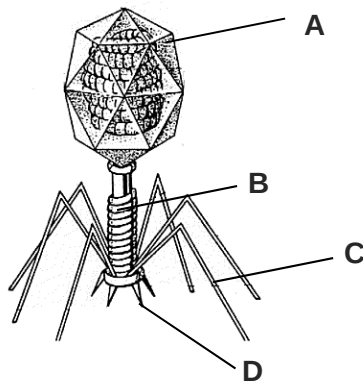
This document consists of **22** printed pages.

- 1 A severe inherited condition arises from the failure to produce an enzyme that breaks down glycoproteins in cells. The condition can be diagnosed from an electron micrograph of a patient's cells.

Which abnormality would be observed in these cells?

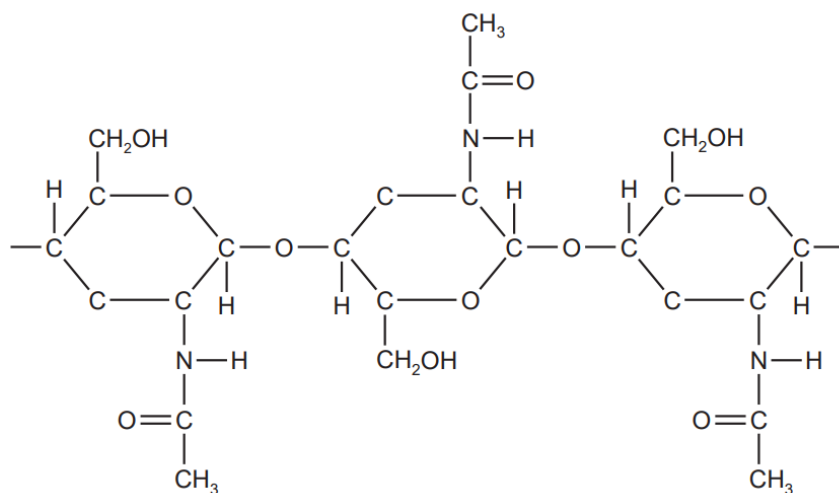
- A an incomplete chromosome due to lack of a gene
- B larger lysosomes due to accumulation of glycoprotein
- C less endoplasmic reticulum due to a reduction in protein synthesis
- D thinner cell surface membrane due to lack of glycoprotein

- 2 The diagram shows a bacteriophage.



Which structure plays the most important role in host recognition?

- 3 The diagram shows the structure of the polysaccharide chitin which is found in the cell wall of fungi.



Which statements are correct for chitin and for cellulose?

- 1 The monomers are joined by 1, 4 glycosidic bonds.
 - 2 Every second monosaccharide in the polysaccharide chain is rotated by 180°.
 - 3 The polysaccharide contains the elements carbon, hydrogen, oxygen and nitrogen.
- A** 1, 2 and 3
- B** 1 and 2 only
- C** 1 and 3 only
- D** 2 only

- 4 The cells in the roots of beetroot plants contain a red pigment.

When pieces of root tissue are soaked in cold water, some of the red pigment leaks out of the cells into the water.

An experiment was carried out to investigate the effect of temperature on the loss of red pigment from the root cells. It was found that the higher the temperature of the water, the higher the rate of loss of red pigment from the root cells.

Which of these statements could explain this trend?

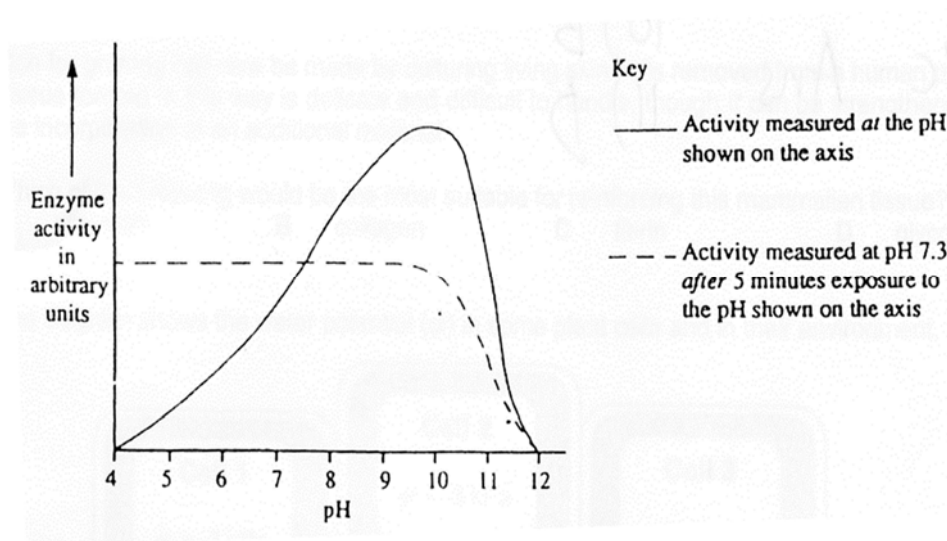
- 1 Enzymes in the cells denature as the temperature increases, so the pigment can no longer be used for reactions inside the cells and diffuses out.
- 2 As the temperature increases, the tertiary structure of protein molecules in the cell surface membrane changes, increasing the permeability of the membrane.
- 3 Phospholipid molecules gain kinetic energy as temperature rises, increasing the fluidity of the phospholipid bilayer and allowing pigment molecules to diffuse out more easily.

- A 1 and 2
- B 2 and 3
- C 2 only
- D 3 only
- 5 The R group of the amino acid serine is $-\text{CH}_2-\text{OH}$. The R group of the amino acid alanine is $-\text{CH}_3$. Where would you expect to find these amino acids in a globular protein soluble in aqueous solution?
- A Serine would be in the interior, and alanine would be on the exterior of the globular protein.
- B Alanine would be in the interior, and serine would be on the exterior of the globular protein.
- C Both serine and alanine would be in the interior of the globular protein.
- D Both serine and alanine would be in the interior and on the exterior of the globular protein.

6 Which row correctly describes the structure of collagen?

A	covalent bonds hold the polypeptides within the triple helices together	about one third of the amino acids in a molecule are glycine	collagen does not have a quaternary structure
B	each of the three polypeptide strands forms a right-handed helix	there is a high proportion of the amino acids proline and glycine	the triple helices are insoluble in water
C	the polypeptides in a triple helix are held together by hydrogen bonds	the triple helices are cross bonded to one another by hydrogen bonds	the glycine side chains are always on the outside of the helix
D	three polypeptide helices are twisted together into a right-handed triple helix	triple helices cross bond to one another with staggered ends	every third amino acid in a polypeptide is usually glycine

7 The graph shows the effects of pH on the activity of the enzyme monoamine oxidase, as measured by two different methods.



Which hypothesis is not supported by these results?

- A** Both acid and strong alkali denature the enzyme.
- B** The optimum pH is alkaline.
- C** Strong alkali causes a reversible change in the tertiary structure of the enzyme.
- D** The change in the catalytic properties of the enzyme caused by acid is reversible.

- 8 Following a heart attack, the enzyme lactate dehydrogenase leaks into the blood plasma from damaged heart muscle.

Which steps are required to obtain the best estimate of lactate dehydrogenase activity in a sample of blood plasma?

	sterilise blood plasma by heating	incubate with substrate for lactate dehydrogenase	incubate with lactate dehydrogenase inhibitor
A	x	✓	x
B	✓	x	✓
C	x	✓	✓
D	✓	✓	✓

key

✓ = step required

x = step not required

- 9 Stem cells are found in many tissues that require frequent cell replacement such as the skin, the intestine or the blood.

However, within their own environments, a bone marrow cell cannot be induced to produce a skin cell and a skin cell cannot be induced to produce a bone marrow cell.

Which statement explains this?

- A** Different stem cells have only the genes required for their particular cell line.
- B** Genes not required for a particular cell line are methylated.
- C** Genes not required for a particular cell line are removed using restriction enzymes.
- D** mRNA that is not required for a particular cell line is destroyed.

10 Which statements concerning DNA and RNA are correct?

- 1 Adenine and guanine are bases that have a double ring structure; cytosine, thymine and uracil are bases with a single ring structure.
- 2 An adenine nucleotide from DNA is the same as an adenine nucleotide from RNA; DNA adenine pairs with thymine and RNA adenine pairs with uracil.
- 3 The base pairing that occurs in a double DNA helix and when RNA is synthesised during transcription is always according to the rule that a purine pairs with a pyrimidine.
- 4 The two polynucleotides on a DNA molecule run in opposite directions so that the double helix formed has two strands that are parallel to each other.

- A** 1, 2 and 3
B 1, 2 and 4
C 1, 3 and 4
D 2, 3 and 4

11 The table compares the structure and function of some elements involved in transcription.

	made of protein	interacts with protein	codes for protein
RNA polymerase	1	2	3
promoter	4	5	6
terminator	7	8	9
gene	10	11	12

Which combination of numbers link the four elements listed to their structures and functions?

- A** 1, 5, 6, 9 and 12
B 1, 5, 8, 11 and 12
C 2, 6, 7, 8 and 11
D 3, 4, 8, 10 and 12

12 Which statements about bacterial genetic transfer are incorrect?

- 1 In transformation, bacterial cells which possess competence factors can take up only plasmids from the surroundings.
- 2 Homologous recombination is always involved in bacterial genetic transfer.
- 3 After conjugation, the donor and recipient cells always contain the same genetic information.
- 4 Binary fission will not contribute to genetic variation in bacterial chromosomes.

- A** 1, 2 and 3
B 1, 2 and 4
C 1 and 3 only
D 2 and 3 only

13 In which process do nucleosomes play a role in eukaryotes?

- 1 tRNA activation
- 2 transcription regulation
- 3 DNA supercoiling

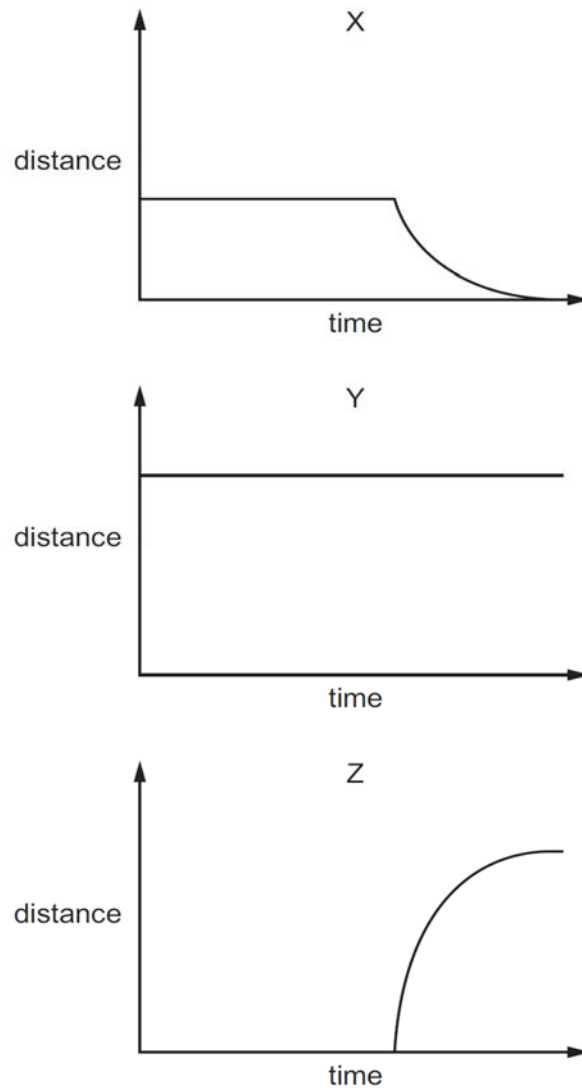
- A** 1, 2 and 3
B 2 and 3 only
C 2 only
D 3 only

14 A mutation occurred in an *Escherichia coli* cell. Given that glucose and lactose are both absent from the growth medium, and the structural genes of the *lac* operon are expressed efficiently, where could the mutation have taken place in the *E. coli* cell?

- 1 in the operator of the *lac* operon
- 2 in the *lacI* regulatory gene
- 3 in the promoter of the *lac* structural genes

- A** 1, 2 and 3
- B** 1 and 2 only
- C** 2 and 3 only
- D** 1 only

- 15 The graphs show various distance measurements taken from metaphase of mitosis onwards. The graphs are to scale when compared to one another.



Which row correctly identifies the distance measurement for each graph?

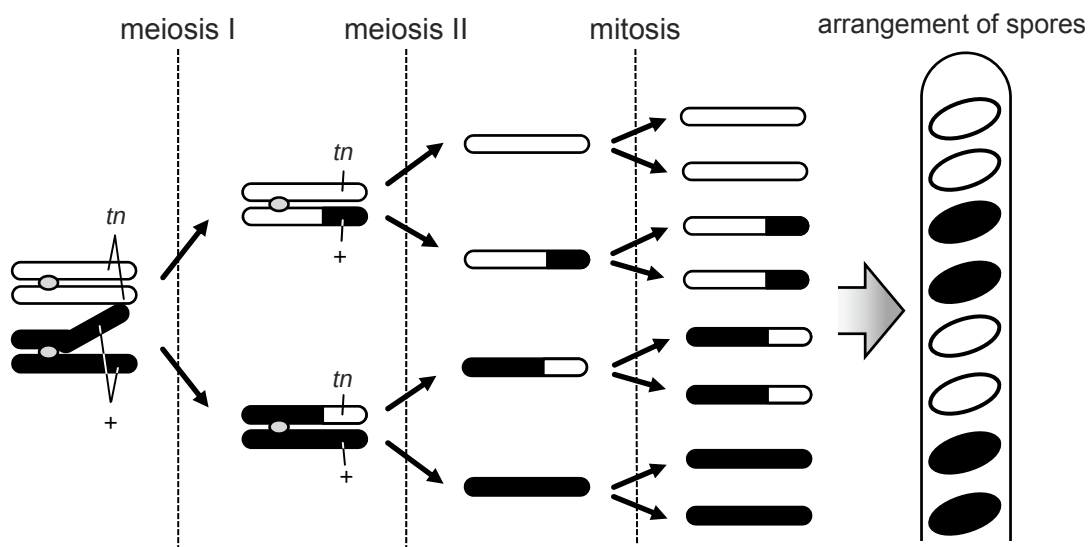
	X	Y	Z
A	distance between poles of spindle	distance between sister chromatids	distance of centromeres from poles of spindle
B	distance between poles of spindle	distance of centromeres from poles of spindle	distance between sister chromatids
C	distance of centromeres from poles of spindle	distance between poles of spindle	distance between sister chromatids
D	distance of centromeres from poles of spindle	distance between sister chromatids	distance between poles of spindle

16 Which feature of the life cycle of some viruses may result in the development of cancer?

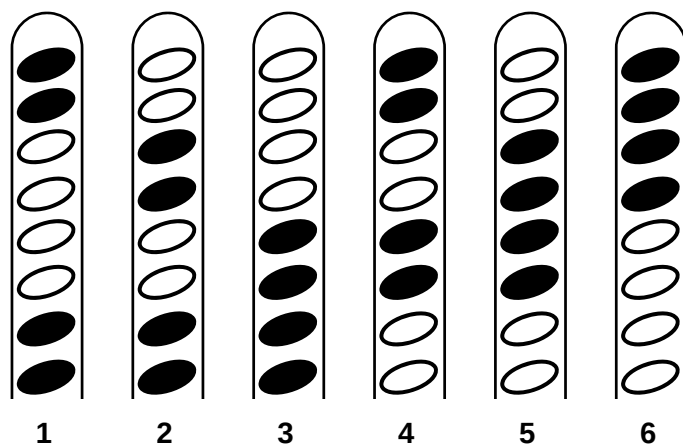
- A** Viral RNA can integrate into the chromosomes of host cells.
- B** Viruses can cause cell lysis and spread to other host cells.
- C** Viruses can cause loss of function mutations in proto-oncogenes.
- D** Viruses can increase the rate of the cell cycle of host cells.

- 17 Two strains of a certain mould, one that produced black spores (+) and the other that produced white spores (*tn*) were crossed.

After fertilisation, the resulting hybrid mould formed four spores by meiosis. Each of these spores then divided once by mitosis so that eight spores were formed in a row. The relative arrangements of the black and white spores made it possible to identify the positions of the planes of the first and second meiotic divisions.



Random examination of the rows of spores formed revealed six possible patterns as shown below.



In which two of these did crossing over not occur?

- A 1 and 4
- B 2 and 4
- C 3 and 5
- D 3 and 6

- 18** In a monohybrid, sex-linked genetic cross involving dominant and recessive alleles, different phenotypes are observed.

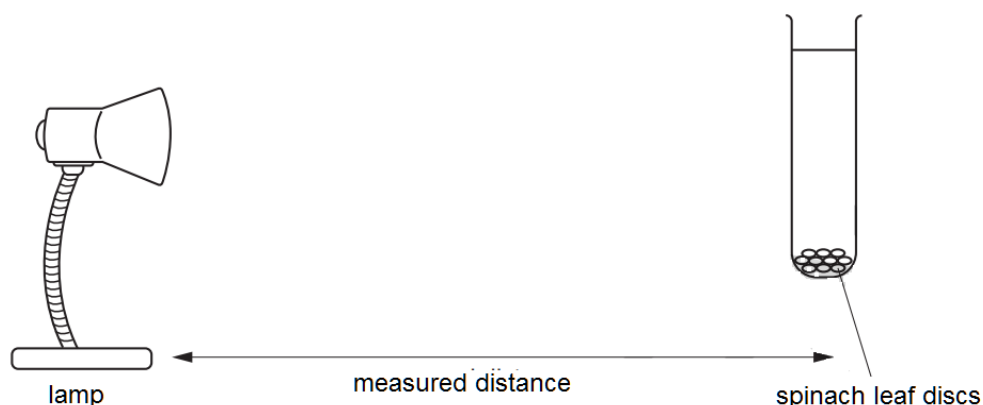
Which statement correctly explains these different phenotypes?

- A** Expression of the recessive allele only occurs in males, because the Y chromosome lacks the relevant gene.
 - B** The nucleotide sequences of the two alleles each produces a different mRNA molecule, only one of which is translated into a functional protein.
 - C** The recessive allele present in the male is unlikely to be transcribed and translated, as the male does not have a corresponding nucleotide sequence on the shorter Y chromosome.
 - D** Transcription of alleles at different loci leads to transcription and translation of active and inactive enzymes.
- 19** The genetic determination of dogs' coats can be quite complex, with many different genes acting at the same time.
- The dominant allele E gives brown tones. The recessive allele e results in red tones.
 - The colour intensity is due to another gene. The dominant allele B gives a dark colour, whereas the recessive allele b results in a light colour.

What would be the genotype of a light brown dog produced from a cross between a dark brown dog and a light red dog?

- A** EEbb
- B** EeBb
- C** eeBb
- D** Eebb

- 20 A student carried out an investigation into the effect of light intensity on photosynthesis. Several groups of spinach leaf discs were placed in test tubes of water. The discs all sank to the bottoms of the tubes. Each tube was placed at a measured distance from a lamp.



As photosynthesis occurs, the build-up of oxygen gas in the leaf discs causes them to rise from the bottom of the tube upwards. The results are shown in the table below.

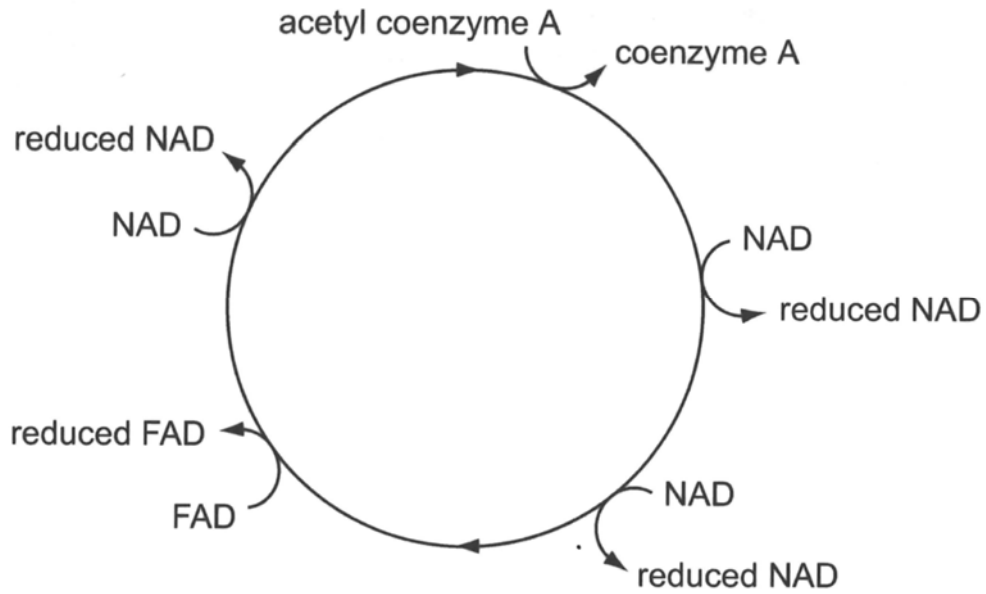
tube number	distance from lamp / mm	time taken for five discs to float / s
1	50	125
2	100	210
3	150	360
4	200	600
5	250	none floated in the time available

Which of these statements are true?

- 1 The compensation point occurs between 200 and 250 mm.
- 2 A variable which is controlled is the distance of the tube from the light source.
- 3 The time taken for the discs to rise is directly proportional to the distance from the lamp.

- A** 1, 2 and 3
B 1 and 3 only
C 1 only
D 2 and 3 only

21 The diagram shows the reactions of the hydrogen carriers in the Krebs cycle.

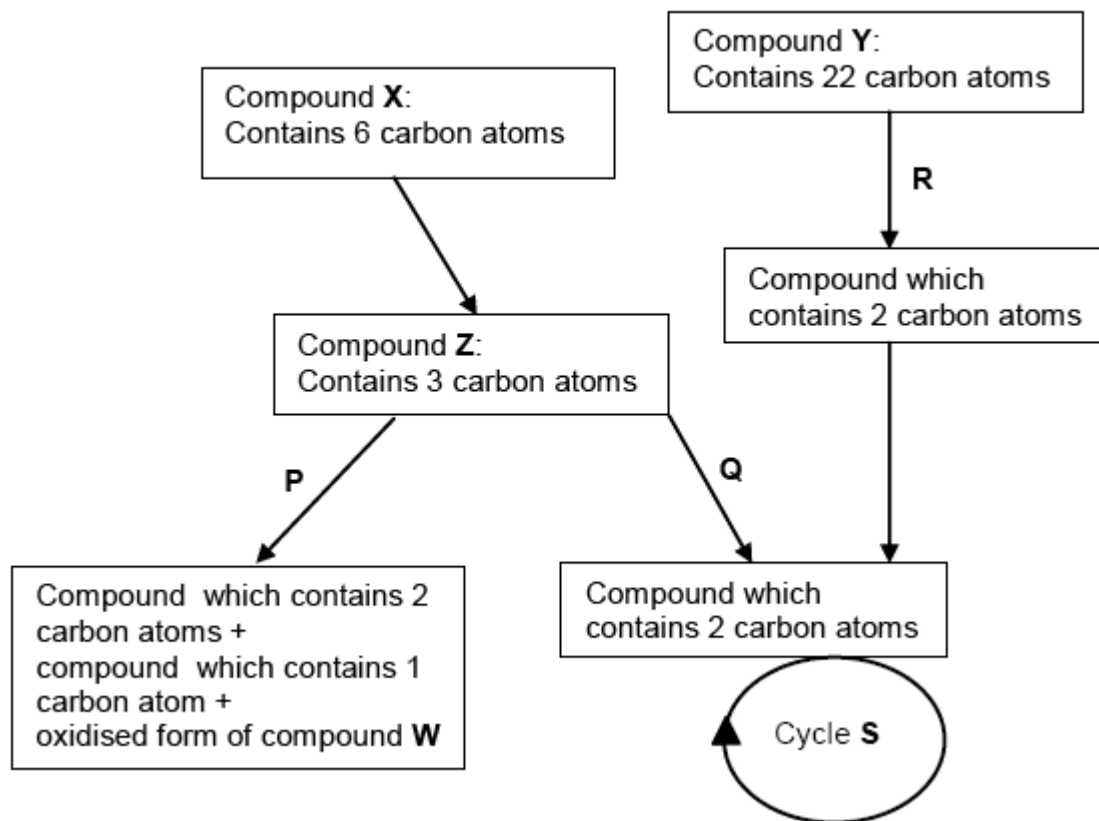


The average yield of ATP, in oxidative phosphorylation, is 2.5 molecules from each molecule of reduced NAD and 1.5 molecules from each molecule of reduced FAD.

What is the average yield of ATP from the hydrogen carriers reduced in the Krebs cycle from one molecule of glucose?

- A 9
- B 18
- C 28
- D 32

- 22 The diagram shows the flow of carbon atoms in cellular respiration in a plant cell, where processes **P**, **Q** and **R** are key stages.



Which statements are correct?

- 1 Process P involves the formation of lactic acid and regeneration of the oxidised form of compound W.
- 2 Compound Y is starch which hydrolyses into acetyl-CoA.
- 3 In process Q, compound Z undergoes oxidative decarboxylation.
- 4 Cycle S produces large amounts of reduced coenzymes for oxidative phosphorylation.
- 5 One molecule of compound X yields 36-38 ATP molecules when completely oxidised.

- A** 1, 2 and 4
B 1 and 3
C 2 and 5
D 3, 4 and 5

23 What is the correct sequence of events in a cell in response to insulin?

- A** Hormone binding to receptor → dimerisation of polypeptides → cross phosphorylation → activation of insulin response protein → activation of glycogen phosphorylase
- B** Hormone binding to receptor → dimerisation of polypeptides → cross phosphorylation → activation of insulin response protein → activation of glycogen synthase
- C** Hormone binding to receptor → cross phosphorylation → dimerisation of polypeptides → activation of protein kinases → activation of glycogen phosphorylase
- D** Hormone binding to receptor → cross phosphorylation → dimerisation of polypeptides → activation of insulin response protein → activation of glycogen synthase

24 Animals with horizontal stripes are bitten less frequently by tsetse flies. The flies carry diseases that infect zebras.

Which explains how zebras might evolve to have more horizontal stripes?

- A** Bites from tsetse flies cause mutations. If a zebra has a mutation it will die and not pass its genes to its offspring which will not have more horizontal stripes.
- B** If two zebras with horizontal stripes mate, their offspring will have more horizontal stripes. Horizontal stripes will become dominant. This is natural selection.
- C** Tsetse flies are a selection pressure. The zebras would gradually develop more horizontal stripes and pass them on to their offspring so they are not bitten by flies.
- D** Zebras with more horizontal stripes get fewer diseases from tsetse flies. These zebras live longer and breed more, passing the allele for more horizontal stripes to their offspring.

- 25** Two areas of molecular biology that have received considerable attention in evolutionary studies are the genetic code and cytochrome *c*. Cytochrome *c* is an essential component of all respiratory electron transport chains.

Which statements lend evidence to the ideas that

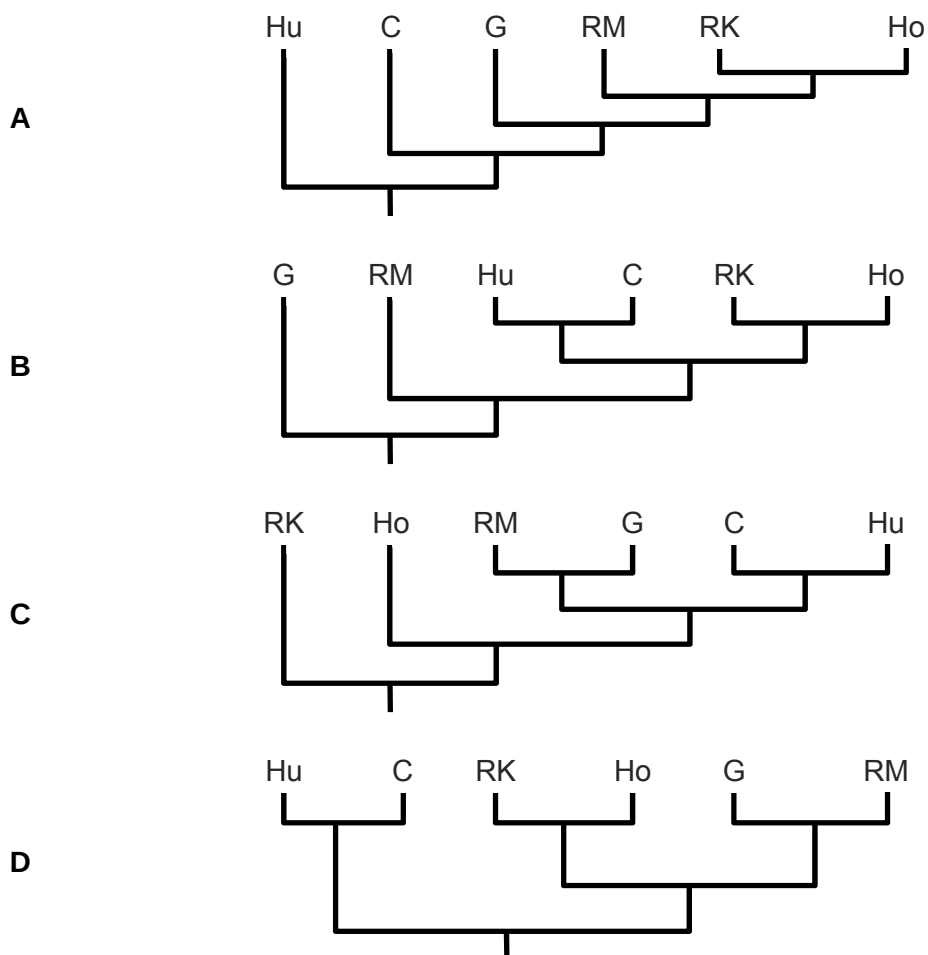
- all living organisms are related
 - there is a single, rather than a multiple, origin of life?
- 1 The almost universal nature of the genetic code is a result of evolutionary convergence from multiple lineages.
 - 2 The sequence of the amino acids in cytochrome *c* is similar in organisms that are from similar environments or with similar metabolic demands.
 - 3 The majority of organisms have the same, or similar, amino acid sequences for cytochrome *c*.
 - 4 When transferred into a very dissimilar organism, a gene coding for cytochrome *c* will lead to the expression of a protein that will function in the other organism.

- A** 1 and 2
- B** 1, 3 and 4
- C** 2 and 3
- D** 3 and 4

- 26 The table below shows the amino acid sequence of part of the β -haemoglobin molecule found in six different vertebrates. The β -globin chain has a total of 146 amino acids. Only the amino acids found in positions 101 to 116 are represented.

vertebrate	amino acid position															
	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116
human (Hu)	glu	asn	phe	arg	leu	leu	gly	asn	val	leu	val	cys	val	leu	ala	his
red kangaroo (RK)	glu	asn	phe	lys	leu	leu	gly	asn	ile	ile	val	lle	cys	leu	ala	glu
rhesus monkey (RM)	glu	asn	phe	lys	leu	leu	gly	asn	val	leu	val	cys	val	leu	ala	his
gorilla (G)	glu	asn	phe	lys	leu	leu	gly	asn	val	leu	val	cys	val	leu	ala	his
chimpanzee (C)	glu	asn	phe	arg	leu	leu	gly	asn	val	leu	val	cys	val	leu	ala	his
horse (Ho)	glu	asn	phe	arg	leu	leu	gly	asn	val	leu	ala	leu	val	val	ala	arg

Using the information above, a suitable phylogenetic diagram would be

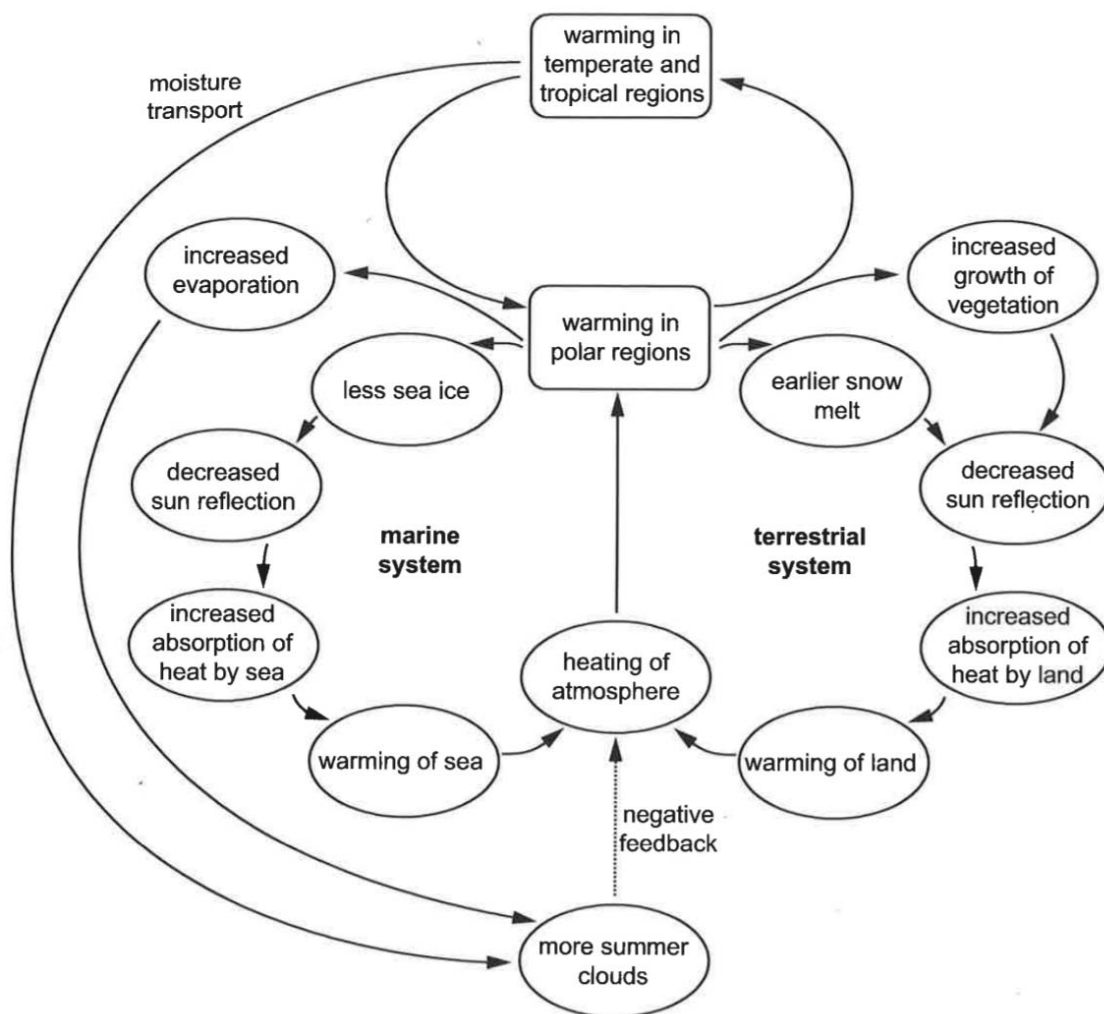


- 27** Which statement regarding a B cell expressing both IgM and IgD on its membrane is incorrect?
- A** The L chains of the IgM and IgD have identical amino acid sequences.
 - B** The constant parts of the H chains of the IgM and IgD have different amino acid sequences.
 - C** The IgM and IgD have different antigenic specificities.
 - D** If it is triggered by antigen and T-cell signals to proliferate and differentiate, it may differentiate into a plasma cell that may secrete IgG, IgE, or IgA antibodies.
- 28** After contact with the polio virus a child developed polio and recovered. Twelve months later the child came in contact with the polio virus again but did not show any symptoms of the disease.

This happened because, shortly after the first infection, the child

- A** had an injection of polio antibodies.
- B** grew B memory cells specific to polio.
- C** grew B memory cells that could respond to any virus.
- D** developed T cells that consumed the new polio particles.

- 29 The diagram shows the effect of increasing temperatures on the ice and snow cover at the polar regions.



Which effect of higher temperatures in the polar regions could increase global warming?

- A Increased evaporation leads to more rainfall, which absorbs heat from the land and the sea.
- B Melting of ice and snow results in less reflection of sunlight and more heat absorption by the Earth.
- C Melting of sea ice caused more cloud formation, which increases absorption of heat in the atmosphere.
- D Earlier melting of snow allows vegetation cover to increase faster, reducing loss of heat from the surface of the Earth.

- 30** Rice crops in Japan are damaged by the green rice leafhopper (*Nephotettix cincticeps*), a pest that reduces crop yield.

In a study of the effect of climate change on crop damage by the green rice leafhopper, it was found that an increase in winter temperatures caused an increase in crop damage, while an increase in summer temperatures caused a decrease in crop damage.

Which of the following are possible explanations for these findings?

- 1 Increased temperatures in the summer cause a rise in metabolic rate that results in the pests reproducing more rapidly.
 - 2 Increased temperatures in the summer raise the metabolic rate above the range that the pests can tolerate.
 - 3 Increased temperatures in the winter disrupt the pests' life cycle and result in fewer being able to reproduce.
 - 4 Increased temperatures in the winter allow more pests to survive and results in an increase in the pest population.
- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4